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Title

**Assessment of Alzheimer's Disease
Progression Using Comorbidity-Aware
Meta-Bayesian Framework with
Resting-State fMRI**

REVIEW REPORT – II

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Abstract

Alzheimer’s disease (AD) progresses through intermediate stages that are difficult to separate using clinical assessment alone, and resting-state functional MRI (rs-fMRI) offers a non-invasive window onto the functional disruption that precedes structural change. This project develops a staging framework that learns a shared, biomarker-invariant representation of AD progression across multiple rs-fMRI biomarkers while accounting for patient comorbidity and prediction uncertainty.

The work reported here establishes the complete data foundation of that framework. A cohort of 175 ADNI rs-fMRI acquisitions was audited; 103 acquisitions initially had metadata but no image data, of which 29 were recovered from raw DICOM archives, and two were excluded for documented data-quality reasons. An exhaustive audit of all 175 JSON sidecars and representative DICOM headers across 17 protocol signatures established that slice-timing information is not recoverable for this dataset, so slice-timing correction was omitted rather than estimated from unsupported assumptions. Because no T1-weighted anatomical image exists for any subject, a direct EPI-to-MNI normalisation route was adopted using symmetric diffeomorphic registration, followed by 4 mm resampling, 6 mm smoothing, linear detrending, a 27-regressor nuisance model and 0.01–0.10 Hz band-pass filtering. Of 173 eligible acquisitions, 165 were preprocessed successfully; the eight failures share a single root cause, a 6.02 s repetition time whose Nyquist limit invalidates the fixed filter band.

All 165 acquisitions were then parcellated into the 246-region Brainnetome atlas, achieving complete 246/246 regional coverage, and four biomarkers — ALFF, ReHo, functional connectivity and degree centrality — were generated with per-acquisition cross-file consistency verification. Degree centrality recomputed independently from every stored connectivity matrix agreed exactly. A leakage-free within-subject normalisation stage was subsequently applied to all 165 acquisitions.

Two findings shaped the design. FC strength, one of the four planned regional biomarkers, is exactly DC/245 and was removed, leaving a 246×3 regional matrix plus the full 246×246 Fisher- z connectivity matrix. Stage-wise group-mean heatmaps over 127 subjects show no univariate progression signal (1, 0 and 1 of 246 regions monotonic, against about 0.7 expected by chance), which motivates a multivariate model. Preprocessing and feature extraction were scheduled for Reviews II and III in the Review I plan and are both complete, so the data side is one milestone ahead. The modelling stages have not been started and no performance result is reported.

Keywords: Alzheimer’s disease; resting-state fMRI; Brainnetome atlas; functional connectivity; ALFF; ReHo; disease progression staging

1. Introduction

Alzheimer’s disease is a progressive neurodegenerative disorder and the leading cause of dementia worldwide. Following the staging scheme fixed at Review I, its clinical course is divided into six ordered stages: cognitively normal (CN), subjective memory concern (SMC), early mild cognitive impairment (EMCI), mild cognitive impairment (MCI), late mild cognitive impairment (LMCI) and Alzheimer’s disease (AD). The boundaries between these intermediate stages are diagnostically important, because therapeutic and care decisions differ substantially between them, yet they are difficult to resolve from cognitive testing alone. Distinguishing them reliably from non-invasive imaging is therefore of direct clinical value.

Resting-state functional MRI measures spontaneous low-frequency fluctuations in the blood-oxygen-level-dependent (BOLD) signal while the subject is at rest. Because functional disruption in AD is believed to precede measurable structural atrophy, rs-fMRI derived measures are attractive candidate markers of early progression. Several complementary families of rs-fMRI measure exist. Amplitude of low-frequency fluctuation (ALFF) quantifies the local strength of spontaneous activity [5]; regional homogeneity (ReHo) quantifies the local temporal coherence of neighbouring voxels [4]; functional connectivity (FC) quantifies statistical dependence between distant regions; and degree centrality (DC) summarises how strongly a region participates in the whole-brain network [7]. Each captures a different aspect of the same underlying physiology.

This diversity is both an opportunity and a difficulty. Individual biomarkers are known to be noisy and to vary with acquisition protocol, so a model trained on one alone tends to generalise poorly. The central premise of this project is that a representation shared across several biomarkers — one that is invariant to which biomarker it was derived from — should be more stable than any single measure. A further complication is that AD rarely occurs in isolation: age, APOE4 genotype, depression, hypertension and diabetes all modulate both brain function and cognitive decline, and a model that ignores them attributes comorbidity-driven variation to disease stage. Finally, because staging decisions are clinical, a point prediction is insufficient; a calibrated measure of uncertainty is required for a prediction to be actionable.

The significance of the present stage of work lies in the data foundation. Every downstream modelling claim rests on the correctness of the preprocessing and biomarker extraction beneath it, and on an honest account of what the source data can and cannot support. Accordingly, this report emphasises verified, reproducible processing with explicit documentation of every point at which the dataset was found

to be limiting.

1.1 Objectives

1. To audit, recover and organise a usable ADNI rs-fMRI cohort spanning six diagnostic groups, documenting every exclusion and every metadata limitation rather than silently discarding data.
2. To design and implement a reproducible rs-fMRI preprocessing pipeline that is valid under this dataset’s constraints, specifically the complete absence of T1-weighted anatomical images and of recoverable slice-timing metadata.
3. To parcellate the preprocessed data into the 246-region Brainnetome atlas and generate four complementary regional biomarkers (ALFF, ReHo, FC, DC) with independent numerical verification of every derived quantity.
4. To apply a leakage-free within-subject normalisation that removes acquisition-level scaling without using any cross-subject or label information.
5. To implement and evaluate a multi-attention transformer with meta-learning that discovers a shared, biomarker-invariant progression representation.
6. To fuse that representation with comorbidity variables and to classify disease stage with a Bayesian ordinal model that reports prediction uncertainty, comparing the result against single-biomarker and non-comorbidity-aware baselines.

Objectives 1–4 are complete and are the subject of this report. Objectives 5 and 6 have not yet been started and are discussed as future work.

1.2 Position Against the Review I Timeline

The Review I plan placed *Module 2: Preprocessing & Validation* at Review II and *Module 3: Feature Extraction — Brainnetome-246* at Review III. Both are complete at Review II, so on the data side the project stands a full milestone ahead of its own Review I timeline (Table 1.1). Module numbers in that table are those of the Review I plan; Chapter 4 numbers this report’s modules differently.

Table 1.1: Planned (Review I) versus actual progress at Review II.

Review I module	Planned for	Status at Review II
Module 2: Preprocessing & Validation	Review II	Complete. 165 of 173 eligible acquisitions preprocessed; 8 rejected for a 6.02 s TR; per-acquisition QC and independent verification done.
Module 3: Feature Extraction — Brainnetome-246	Review III	Complete, one review early. 165/165 acquisitions parcellated with 246/246 coverage; ALFF, ReHo, FC and DC generated and verified for all 165.
Modules 4–6: transformer, meta-learning, comorbidity fusion, Bayesian ordinal classifier, dashboard	per Review I Gantt	Not started. No model has been trained; no accuracy, AUC or other performance figure exists.

In addition, within-subject normalisation of all 165 acquisitions is complete. The lead should be read narrowly: it applies to the data foundation only. The comorbidity data are not yet cleaned and the cross-validation folds are not yet built (Section 6.5).

2. First Review - Queries and Answers

- **Query 1** — *[To be completed with the query raised by the review panel at Review I.]*
Answer — *[Response.]*
- **Query 2** — *[To be completed.]*
Answer — *[Response.]*

[TO BE COMPLETED BEFORE SUBMISSION.] Fill in each query and answer from the Review I minutes. The queries raised by the panel are not reproduced here because they are not available in the project repository.

3. Problem statement

Problem. Given a single resting-state fMRI acquisition from a subject with no accompanying T1-weighted anatomical scan and no recoverable slice-timing metadata, together with that subject’s comorbidity profile, assign the subject to one of six ordered disease-progression stages and report a calibrated uncertainty for that assignment.

Inputs.

- One 4D rs-fMRI BOLD series per acquisition (ADNI [1, 2], Philips scanners; 159 of 165 acquisitions are $64 \times 64 \times 48$ voxels, 164 of 165 have $TR \approx 3.0$ s and one has 2.25 s; 140 volumes before 5 are discarded).
- Diagnostic group label per subject, taken only from `participants.tsv`.
- Comorbidity variables per subject: age, APOE4 genotype, MMSE, CDR-SB, geriatric depression score, and cardiovascular and endocrine medical history.

Outputs.

- A predicted progression stage on the ordered scale $CN \rightarrow SMC \rightarrow EMCI \rightarrow MCI \rightarrow LMCI \rightarrow AD$.
- A posterior distribution over stages, from which prediction uncertainty is derived.
- Region-level attribution identifying which of the 246 Brainnetome regions and which biomarkers drove the prediction.

Constraints.

- **No anatomical reference.** Zero of the 131 audited subjects have a T1w image, so anatomical normalisation, subject-specific tissue segmentation and contrast-to-noise estimation are all unavailable.
- **No slice-timing metadata.** Slice-timing correction cannot be applied to any acquisition in the dataset.
- **Small effective sample.** 165 acquisitions originate from only 127 distinct subjects, and the smallest group contains 14 subjects. Model capacity and validation design are bounded by this, not by the acquisition count.
- **Single-worker compute.** All processing must run with one worker on a memory-constrained WSL2 environment, so the pipeline must be checkpointed and resumable.

Success criteria.

- *Data stage (this report)*: complete 246/246 regional coverage for every acquisition; zero non-finite values in any derived feature; every derived quantity reproducible by independent recomputation to within 10^{-10} .
- *Model stage (future)*: staging performance exceeding single-biomarker and comorbidity-agnostic baselines under subject-grouped cross-validation, with calibrated uncertainty.

4. Methodology

The project follows a fixed seven-block design. Blocks 1–5 (audit, preprocessing, parcellation, biomarkers, normalisation) are complete; blocks 6–7 (representation learning and the Bayesian ordinal classifier) have not yet been started.

4.1 Proposed Workflow

- Step 1:** Audit the raw cohort, recover acquisitions lacking image data from DICOM, and document all exclusions.
- Step 2:** Preprocess each acquisition into a common template space under the dataset’s metadata constraints.
- Step 3:** Parcellate into the Brainnetome-246 atlas and extract regional time series.
- Step 4:** Generate ALFF, ReHo, FC and DC, and verify every derived quantity.
- Step 5:** Normalise within subject, without using cross-subject or label information.
- Step 6:** Learn a shared biomarker-invariant representation using a multi-attention transformer with meta-learning.
- Step 7:** Fuse comorbidity information and classify stage with a Bayesian ordinal model that reports uncertainty.

4.2 Deviations from the Review I Plan

Auditing the data changed three specifications made at Review I. Each change was forced by what the dataset contains, and none was made for convenience (Table 4.1).

Table 4.1: Deviations from the Review I plan and their evidence.

Review I specified	Implemented	Evidence and consequence
Slice-timing correction (§5.2.2)	Step omitted; recorded as NOT_PERFORMED in every provenance file	No <code>SliceTiming</code> field in any of 175 JSON sidecars; DICOM headers from 17 protocol signatures, including private vendor tags, give volume-level timing only. Outcome <code>SLICE_TIMING_NOT_RECOVERABLE</code> . The step was omitted, not assumed away, and no timing was inferred.
Co-registration to structural scans (§5.2.2)	Direct EPI-to-MNI registration (ANTsPy SyN) with population-template tissue priors	Zero of 131 subjects have a T1w image. Subject segmentation is therefore impossible; CNR is not computable for any acquisition, and the CSF regressor failed its reliability gate in all 165 (Section 5.2).
246×4 regional feature matrix	246×3 regional matrix plus the full 246×246 Fisher- z FC matrix	FC strength = DC/245 exactly, so one column was a rescaling of another (Section 5.6). FC is not dropped; it moves to its full matrix form.

4.3 Architectural Design for Proposed System

Figure 4.1 shows the system as implemented. Data flows strictly left to right; each stage reads only the outputs of the stage before it and writes to its own directory, so no stage can modify its own input. This separation is what makes each stage independently auditable.

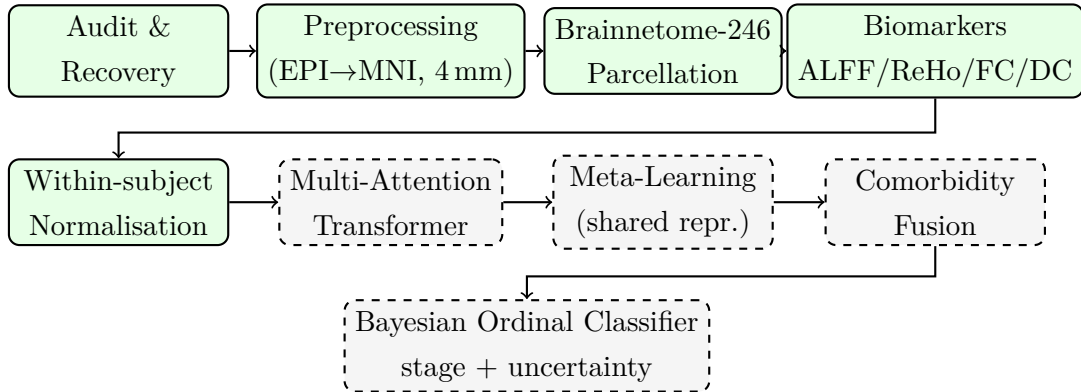


Figure 4.1: High-level architecture of the proposed system. Solid green blocks are implemented and verified; dashed grey blocks are planned and not yet started.

4.4 Modules

4.4.1 Module 1: Data Audit, Recovery and Organisation

Purpose. Establish a trustworthy cohort and an explicit record of what the source data does and does not contain. **Inputs.** Raw ADNI DICOM archives and partially converted NIfTI files. **Processing.** Inventory every acquisition; locate acquisitions with metadata but no image data and convert them from DICOM; detect duplicate and truncated acquisitions by checksum and header inspection; audit slice-timing recoverability; organise into a BIDS-like layout with group membership taken only from `participants.tsv`. **Outputs.** Organised cohort, exclusion list, and audit reports. **Dependencies.** None.

4.4.2 Module 2: Preprocessing

Purpose. Bring every acquisition into a common template space with nuisance signal suppressed. **Inputs.** Module 1 cohort. **Processing.** The eleven steps of Table 5.1. **Outputs.** One preprocessed 4D BOLD volume, brain mask, motion parameter file, nuisance design matrix, provenance record and QC report per acquisition. **Dependencies.** Module 1.

4.4.3 Module 3: Parcellation and Regional Time-Series Extraction

Purpose. Reduce voxel-level data to a fixed 246-region representation. **Inputs.** Preprocessed BOLD; original 2 mm Brainnetome atlas. **Processing.** Reorient the atlas losslessly, resample it onto each subject’s own BOLD grid using nearest-neighbour interpolation only, validate regional coverage, and extract the mean BOLD time course of each region. **Outputs.** Subject-aligned atlas, $T \times 246$ regional time-series matrix, regional voxel counts and QC. **Dependencies.** Module 2.

4.4.4 Module 4: Biomarker Generation

Purpose. Derive the four complementary biomarkers. **Inputs.** Preprocessed BOLD, aligned atlas, regional time series. **Processing.** ALFF and ReHo are computed voxel-wise and then averaged within each region; FC is computed between all region pairs; DC is derived from FC. Definitions are given in Table 5.2. **Outputs.** Per-acquisition biomarker vectors, the full connectivity matrix, a combined regional feature matrix, and QC. **Dependencies.** Module 3.

4.4.5 Module 5: Within-Subject Normalisation

Purpose. Remove acquisition-level scaling that would otherwise dominate cross-subject comparison. **Inputs.** Module 4 biomarkers. **Processing.** Every statistic is computed from the acquisition’s own 246 values, so the stage is leakage-free by construction. Methods are given in Table 5.3. **Outputs.** Normalised features in a separate directory tree; raw biomarkers are never modified. **Dependencies.** Module 4.

4.4.6 Module 6: Modelling and Evaluation (planned)

Purpose. Learn the shared progression representation, fuse comorbidity, and classify stage with uncertainty. **Inputs.** Module 5 features; cleaned comorbidity table. **Processing.** Subject-grouped stratified cross-validation, with cross-subject scaling fitted on training folds only; multi-attention transformer; meta-learning across biomarkers; comorbidity fusion; Bayesian ordinal classification; attention-based explanation. **Outputs.** Stage predictions, uncertainty, region and biomarker attribution. **Dependencies.** Modules 5 and 1.

5. Implementation

All processing ran under WSL2 (Ubuntu) on Windows 11 with Python 3.12.3, NumPy 2.3.5, SciPy 1.15.3, NiBabel 5.4.2, Nilearn 0.14.1, ANTsPy 0.6.3, TemplateFlow 25.1.2 and FreeSurfer 7.4.1. Every long-running job used a single worker, with all thread-count environment variables pinned to one.

5.1 Module 1: Data Audit, Recovery and Organisation

The initial audit covered 131 subjects and found 72 valid NIfTI files against 103 records that had JSON metadata but no image data at the audited location. Of these, 29 were recovered by converting raw DICOM archives (9 in the MCI group and 20 in the SMC group, the latter representing 62.5% of that group’s final data), with no conversion failures; the remaining records were reconciled against NIfTI files found by the later per-group mapping audits. A dataset-wide checksum audit of 175 image files found no orphans and exactly one hash collision, a subject exported twice under different series numbers; the duplicate was excluded and the original retained. A second acquisition was excluded for containing only 7 volumes where all others contain 140, a truncation traced to the source export rather than to any processing step. Both exclusions are recorded in a canonical exclusion list that every batch script consults.

An early working note recorded six AD acquisitions as unrecoverable. This figure was re-checked directly against the current source data before the production run and found to be stale: all 20 AD subjects and all 25 AD acquisitions were present and valid, and all were processed.

The slice-timing audit examined all 175 JSON sidecars and representative DICOM headers spanning all 17 distinct protocol signatures in the archive, inspecting private vendor groups as well as standard tags. No sidecar contained a `SliceTiming` field. Within the DICOM headers, `AcquisitionTime` and `TriggerTime` were identical across all slices sampled within a volume, confirming volume-level rather than slice-level timing granularity. A Philips private tag provides a per-slice *spatial* index, but no tag anywhere states the temporal acquisition order, and the spatial index was deliberately not used to infer it. No multiband tag was found among decodable names. The audit’s conclusion was recorded as `SLICE_TIMING_NOT_RECOVERABLE`, and no timing value was inferred or fabricated at any point.

5.2 Module 2: Preprocessing

Table 5.1: Implemented preprocessing pipeline.

Step	Implementation
Input validation	4D shape, all values finite, non-degenerate affine, $TR > 0$, non-empty image
Volume removal	First 5 volumes discarded for signal equilibration
Slice timing	Not performed — not recoverable for this dataset
Motion correction	FreeSurfer [11] FS-FAST <code>mc-afni2</code> (AFNI [12] <code>3dvolreg</code>), rigid 6-DOF, reference = first retained frame; native <code>.mcdat</code> preserved
Normalisation	Direct EPI-to-MNI152NLin6Asym via ANTsPy symmetric diffeomorphic registration (<code>SyN</code>) [10]; no T1w exists, so the skull-stripped temporal mean BOLD is registered to the template and the 4D series warped in a single interpolation
Resampling	4 mm isotropic, continuous interpolation
Smoothing	6 mm FWHM Gaussian
Detrending	Per-voxel linear OLS; intercept and linear term removed, temporal mean restored; no quadratic term
Nuisance regression	27 regressors in every acquisition: intercept, Friston-24 [9], white-matter signal from a population template prior, and global signal. A CSF candidate from the same priors failed its reliability gate in all 165 acquisitions
Band-pass filter	0.01–0.10 Hz, 2nd-order Butterworth, zero-phase <code>filtfilt</code> ; Nyquist validity checked per acquisition
QC	tSNR, SNR, framewise displacement [8], DVARS, spatial and temporal entropy

Two implementation points deserve emphasis. First, because no T1w image exists, white-matter and CSF candidate regressors are derived from published population tissue probability priors rather than from subject-specific segmentation; a candidate is included only if its mask retains at least 20 voxels and its correlation with the global signal is below 0.98 in magnitude. The white-matter regressor passed this gate in all 165 acquisitions and the CSF regressor failed it in all 165, so the nuisance design matrix has exactly 27 columns in every acquisition (intercept, 24 Friston terms, white matter, global signal); every provenance record states `CSF_status = NOT PERFORMED`. The ventricular CSF prior is too thin to survive at 4 mm: the corresponding 2 mm pilot mask held only 111 voxels (888 mm^3) after thresholding at 0.90 and intersection with the brain mask, which is roughly 14 voxels on the 4 mm grid, below the 20-

voxel minimum. The voxel count and correlation at the gate were not stored per acquisition, so this is the explanation consistent with the logged evidence rather than a per-acquisition measurement. No substitute CSF regressor was introduced. Second, contrast-to-noise ratio is reported as not reliably computable for all 165 acquisitions, because it requires a tissue segmentation this dataset cannot provide (no T1w image exists for any of the 131 subjects); no segmentation was fabricated to produce a number.

The band-pass step validates, per acquisition, that the requested upper cutoff lies below the Nyquist frequency implied by that acquisition’s own TR. Eight acquisitions failed this check. All eight share a TR of approximately 6.02s, roughly double the 3.0s used elsewhere in the cohort, giving a Nyquist limit near 0.083 Hz — below the 0.10 Hz cutoff. These were refused rather than silently reprocessed with an altered band, since changing the filter for a subset would make those acquisitions non-comparable with the rest. Seven of the eight have a TR of 6.02158s and one has 6.02353s; all eight belong to site 002 (subject identifiers 002_S_XXXX).

5.3 Module 3: Parcellation and Regional Time-Series Extraction

The Brainnetome atlas [3] is supplied at 2 mm in LAS orientation whereas the preprocessed BOLD is RAS, so the atlas is first reoriented to canonical orientation, a lossless array operation involving no interpolation, verified by per-label voxel count equality. It is then resampled onto each subject’s own BOLD grid using *nearest-neighbour interpolation only*. This is essential: atlas values are categorical region identifiers, and linear or spline interpolation would produce fractional labels with no meaning. Regional time series are extracted by direct index masking on the verified-identical grid rather than through a masker utility that might silently re-sample the labels a second time.

5.4 Module 4: Biomarker Generation

Table 5.2: Implemented biomarker definitions.

Biomarker	Definition as implemented	Shape
ALFF	Mean FFT amplitude within 0.01–0.10 Hz of the mean-centred voxel time series, then averaged within each region	(246,)
ReHo	Kendall’s coefficient of concordance over a 27-voxel $3 \times 3 \times 3$ neighbourhood including the centre voxel, with the neighbour count reduced at the mask boundary, then averaged within each region	(246,)
FC	Pearson correlation between all region pairs; signed, unthresholded, diagonal retained	(246, 246)
DC	$DC_i = \sum_{j \neq i} FC_{ij}$; weighted, signed, unthresholded, self-connection excluded	(246,)

ALFF and ReHo are computed at voxel level and only then aggregated to regions, rather than computed from the regional time series, so that local spatial information is used where the definitions require it. The 27-voxel ReHo neighbourhood is the original formulation [4] and the default of established toolboxes, chosen for comparability rather than arbitrarily. Because the preprocessed BOLD has already been band-pass filtered, ALFF is computed on an already band-limited spectrum; this is a documented consequence of the fixed pipeline rather than a change of method.

The connectivity matrix is retained in full. No threshold, binarisation or absolute value is applied, and negative correlations are preserved, since thresholding would impose an unvalidated choice and discard the sign information that distinguishes anti-correlated from unconnected regions.

A single-acquisition pilot was implemented and validated first. Before scaling to the cohort, the production implementation was required to reproduce that pilot: ALFF, ReHo, regional time series, atlas labels and regional voxel counts were recovered bit-identically, and FC and DC agreed to within 1.8×10^{-15} . Only then was the pilot’s own output directory protected and the remaining acquisitions processed.

5.5 Module 5: Within-Subject Normalisation

Table 5.3: Within-subject normalisation, computed from each acquisition’s own 246 regional values only.

Feature	Transform and rationale
mALFF	$\text{ALFF}_i / \overline{\text{ALFF}}$. Raw ALFF is in FFT amplitude units proportional to raw BOLD intensity and spans roughly 8.5×10^2 to 2.2×10^5 across acquisitions; the ratio removes this scaling [6]. A ratio is used rather than a z -score because ALFF is strictly positive
mReHo	$\text{ReHo}_i / \overline{\text{ReHo}}$, removing each subject’s global coherence offset
DC_z	$(\text{DC}_i - \overline{\text{DC}}) / \sigma_{\text{DC}}$ with ddof = 0 [7]. A z -score is used because DC is signed, which rules out min-max scaling
Fisher- z FC	$\text{arctanh}(\text{clip}(r, \pm 0.999999))$ on off-diagonal entries, variance stabilising the correlations. The diagonal is zeroed <i>before</i> the transform so that $\text{arctanh}(1) = \infty$ is never produced, then set to exactly zero; negative correlations stay negative

Leakage-free by construction. Every statistic in Table 5.3 is computed from that acquisition’s own 246 values. No cross-subject statistic and no diagnostic label is used at any point, so the stage cannot carry information from a test subject into training. Cross-subject scaling is deliberately deferred to the modelling stage, where it will be fitted on training folds only. Outputs are written to a separate directory tree, and the script refuses any write path inside a raw `biomarkers` directory.

Table 5.4: Verification of within-subject normalisation (165 acquisitions, 0 failures).

Check	Result
Acquisitions normalised; failures	165 / 165; 0
Non-finite values in inputs; in outputs	0; 0
mean(mALFF) and mean(mReHo) per acquisition	1.000000 in all 165
mean(DC_z), largest magnitude	$\leq 1.6 \times 10^{-16}$
std(DC_z) per acquisition	1.000000 in all 165
Fisher- z diagonal exactly zero	165 / 165
Fisher- z maximum asymmetry	5.329×10^{-15}
Negative correlations preserved as negative	165 / 165
Off-diagonal entries requiring clipping at ± 0.999999	0
Fisher- z range across the dataset	-2.2653 to 2.5924

5.6 Feature Redundancy: FC Strength and DC

Auditing our own features, rather than inheriting them from the design, exposed an exact redundancy. The pipeline stores a regional connectivity strength defined as the *mean* of each region’s off-diagonal connectivity, while DC is the *sum* over the same 245 entries, so

$$\text{FCS}_i = \frac{1}{245} \sum_{j \neq i} \text{FC}_{ij} = \frac{\text{DC}_i}{245}.$$

Table 5.5 gives the measurements.

Table 5.5: Measured equivalence of FC_Strength and DC.

Comparison	Result
Raw outputs, all 165 acquisitions: $\max \text{DC}/245 - \text{FCS} $	6.939×10^{-18}
Raw outputs: minimum Pearson r across acquisitions	1.000000000000
Stage matrices (both within-subject z -scored): $\max \text{DC}_z - \text{FCS}_z $	6.661×10^{-16}
Stage matrices: Pearson r	1.0

Under any scale-invariant normalisation the two columns become numerically identical.

Design correction. Review I specified a 246×4 regional feature matrix (ALFF, ReHo, DC, FC strength). One column was a deterministic rescaling of another, so the independent regional features are three. The corrected model input is a 246×3 regional matrix (mALFF, mReHo, DC_z) *plus* the full 246×246 Fisher- z connectivity matrix. FC is not dropped: it moves from a collapsed per-region scalar into its full matrix form, which carries strictly more information. The raw FC strength files are retained unaltered; only the model feature set changes. This is a correction derived from auditing the features, not a pipeline error.

5.7 Engineering for an Unstable Execution Environment

Early production runs failed repeatedly, and diagnosis of the failure mode materially shaped the implementation. The symptom was that the virtual machine terminated after roughly five to twelve acquisitions with no Python traceback. The cause was sustained small-granularity array input/output against the Windows drive mount, which produced `Errno 5` input/output errors and virtual machine level failures under load. Three changes resolved it. Each acquisition is now staged on the native Linux

filesystem and its results copied to the Windows-visible tree only once complete, so a crash never leaves a partially written output directory. Resume validation reads array shapes through memory mapping and file headers instead of loading every finished acquisition back into memory, which had been re-reading gigabytes on every restart. Finally, a wrapper distinguishes a genuine Python error, which stops the run for inspection, from a virtual machine failure, which triggers a restart and resumption from the last validated checkpoint. Restart count fell from 23 to 2 for the same workload.

Checkpointing re-validates the actual files on disk — shapes, finiteness and structural invariants — before treating any acquisition as complete, so the checkpoint record alone is never trusted.

6. Results and Discussion

6.1 Cohort Composition

Table 6.1: Cohort composition after audit, recovery and exclusion, listed in progression order.

Stage	Eligible	Preprocessed	Biomarkers	Subjects
CN (CN_Final)	29	27	27	26
SMC (SMC_Final)	32	31	31	21
EMCI	30	29	29	26
MCI	35	32	32	14
LMCI	22	21	21	20
AD	25	25	25	20
Total	173	165	165	127

Table 6.1 reports the figure that most constrains the modelling stage. The 165 acquisitions derive from only 127 distinct subjects: 96 subjects contribute one acquisition, 25 contribute two, 5 contribute three and 1 contributes four. The MCI group illustrates why this matters. By acquisition count it is the largest group at 35, but it comprises only 14 subjects, at 2.3 acquisitions per subject. Because validation splits must group by subject — otherwise the same individual appears in both training and test data — 14 is the number that bounds what can be claimed about that class. No subject appears in more than one group, so labels are subject-consistent.

6.2 Preprocessing Quality Control

Table 6.2 summarises acquisition-level quality control by stage. One acquisition is treated separately and is flagged explicitly.

Table 6.2: Preprocessing quality control by stage. All columns except the last exclude the single artifact acquisition described below ($n = 164$ overall). The raw tSNR mean as originally reported by the dataset audit, which includes it, is shown in the last column. FD is mean framewise displacement in mm; “med.” is the median.

Stage	n	tSNR mean	tSNR med.	SNR mean	FD mean	tSNR raw
CN (CN_Final)	26	131.63	132.58	12.95	0.352	12064.36
SMC (SMC_Final)	31	125.71	123.70	13.15	0.375	125.71
EMCI	29	137.49	133.16	13.76	0.355	137.49
MCI	32	137.05	139.24	12.94	0.321	137.05
LMCI	21	127.54	121.41	12.99	0.322	127.54
AD	25	114.06	118.76	13.49	0.439	114.06
All	164	129.40	127.95	13.22	0.360	2082.04

Dataset-wide medians over the 164 acquisitions are tSNR 127.95, SNR 13.12, mean framewise displacement 0.317 mm and raw-intensity DVARS 15270. Contrast-to-noise ratio is not available (N/A) for all 165 acquisitions, since no T1w image exists for any of the 131 subjects. Head motion is substantial: 134 of the 164 acquisitions exceed 0.2 mm mean FD, 90 exceed 0.3 mm and 28 exceed 0.5 mm. High motion is flagged, not excluded, under the frozen processing policy, so motion will enter the modelling stage as a covariate.

Artifact acquisition (reported, not silently dropped). The audit-reported CN tSNR mean of 12064 is produced by a single acquisition, CN_Final/sub-018S4313/ses-01/run-01, whose tSNR is 322,315 against a dataset median of 128. The ratio is mean over standard deviation per voxel, so near-zero-variance voxels inflate it; the project methodology record attributes the value to this effect. The same acquisition has mean FD 1.079 mm (among the four highest of 165) and an anomalously low spatial entropy of 4.21 bits against a cohort mean of 7.18 for the other 164. It is excluded from the tSNR, SNR, FD and DVARS summaries in Table 6.2 and from the medians above, and the raw value is kept visible in the last column. With it removed, the CN tSNR mean is 131.63 (median 132.58), in line with the other stages. The acquisition itself remains among the 165 preprocessed acquisitions and in every downstream count; how it is handled in modelling has not yet been decided.

6.3 Biomarker Quality

Table 6.3: Dataset-level quality control over all 165 processed acquisitions.

Check	Result
Acquisitions with complete biomarker outputs	165 / 165
Processing failures at the biomarker stage	0
Brainnetome coverage 246/246 regions	165 / 165
Regions with zero voxels (any acquisition)	0
Smallest region, across the whole dataset	10 voxels
Regional time-series matrices of shape (135, 246)	165 / 165
Non-finite values in regional time series	0
Regions with zero temporal variance	0
ALFF: non-finite or negative values	0
ReHo: values outside the valid $[0, 1]$ range	0
ReHo range across dataset	0.2352–0.8231
FC: non-finite values	0
FC maximum diagonal error	2.2×10^{-16}
FC maximum asymmetry	2.2×10^{-16}
DC recomputed independently from stored FC, max. difference	0.0×10^0
Acquisitions passing all cross-file consistency checks	165 / 165

The result in Table 6.3 that carries the most evidential weight is the independent recomputation of degree centrality. For every acquisition, DC was recomputed from the stored connectivity matrix in a separate process that read only saved files, and compared against the stored vector; the maximum difference across all 165 acquisitions was exactly zero. This verifies not merely that the files are well formed but that the stored values are the ones the documented definition produces.

Complete 246/246 regional coverage in every acquisition follows from the pipeline design: because all acquisitions are registered to the same template-derived 4mm grid, the aligned atlas is identical across the cohort. This was nonetheless verified per acquisition rather than assumed.

Normalisation was applied to all 165 acquisitions with no failures (Table 5.4). Mean mALFF and mean mReHo equal 1.000000 for every acquisition; DC_z has mean below 1.6×10^{-16} and standard deviation 1.000000; the Fisher- z connectivity matrix is symmetric to 5.3×10^{-15} with a diagonal of exactly zero in all 165 cases. No off-diagonal correlation anywhere in the dataset exceeded the ± 0.999999 clipping bound, so that safeguard never engaged — confirming no region pair is numerically perfectly correlated.

6.4 Stage-wise Group Patterns: A Negative Univariate Result

We tested whether a univariate, per-region group-mean signal distinguishes the six progression stages. Stage-wise matrices of size 246×6 were generated for mALFF, mReHo, DC_z and FC_Strength with subject-level aggregation: acquisitions were averaged within subject first and subjects then averaged within stage, so each of the 127 subjects counts once although there are 165 acquisitions. The subject counts are CN 26, SMC 21, EMCI 26, MCI 14, LMCI 20 and AD 20. Each region was then z -scored across the six stages so that colour shows how a region changes across stages and not how large it is (Figure 6.1). The absolute-value versions (`derivatives/STAGE_HEATMAPS/*_absolute.png`) are available but are not shown, because they are dominated by anatomical differences between regions.

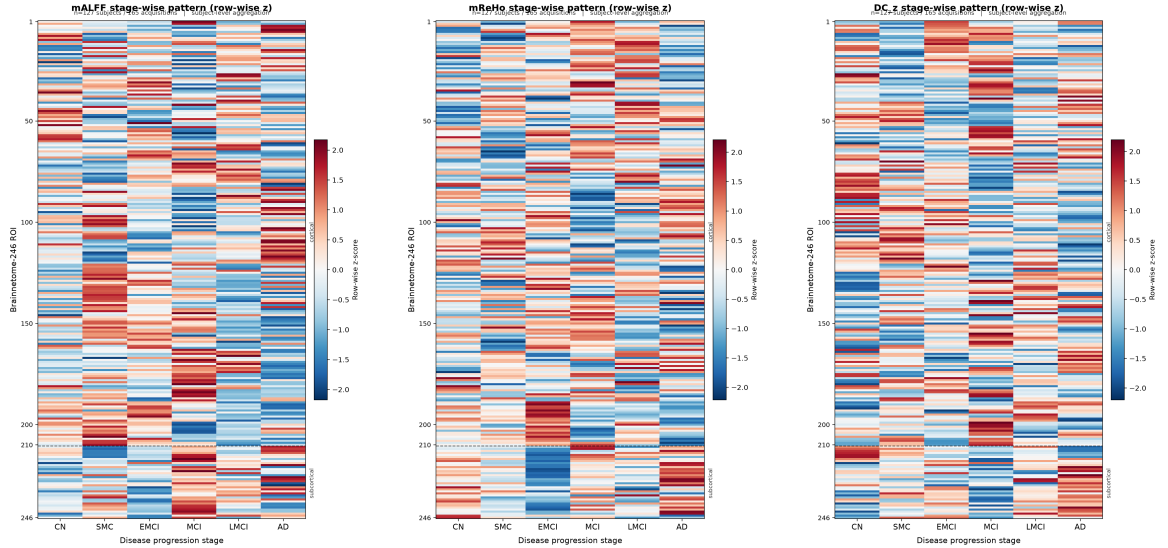


Figure 6.1: Stage-wise regional biomarker patterns across the six-stage progression (CN, SMC, EMCI, MCI, LMCI, AD) for 127 subjects; from left to right mALFF, mReHo and DC_z . Rows are the 246 Brainnetome ROIs in atlas order (cortical 1–210 above the dashed line, subcortical 211–246 below); columns are progression stages. Each ROI is z -scored independently across the six stages, so colour encodes how a region changes across stages rather than its absolute magnitude. Acquisitions were averaged within subject before averaging across subjects, so each of the 127 subjects contributes once. These are descriptive, exploratory visualisations; they do not demonstrate disease progression and no statistical testing was performed.

Table 6.4: Monotonic trends in the stage means. A region counts as monotonic if its six stage means increase, or decrease, strictly from CN to AD. The expectation by chance is $246 \times 2/6! \approx 0.7$, the combinatorial value when the six stage means of a region are exchangeable; it is not a hypothesis test. The last column is the median over regions of the standard deviation of the six stage means, divided by the standard deviation across regions of the per-region mean (population standard deviations).

Biomarker	ROIs with monotonic CN→AD trend	Expected by chance	Stage SD / between-ROI SD
mALFF	1 / 246 (decreasing)	≈ 0.7	0.18
mReHo	0 / 246	≈ 0.7	0.24
DC _z	1 / 246 (increasing)	≈ 0.7	0.23

The result (Table 6.4) is negative and is reported as such, in four steps.

1. We tested for a univariate, per-region group-mean progression signal.
2. It is not detectable. Only 1, 0 and 1 of 246 regions are monotonic from CN to AD, which is no more than the roughly 0.7 that chance alone gives, and the variation across stages is about 4–5 times smaller than the anatomical variation between regions (ratios 0.18–0.24).
3. This is informative, not a failure of the pipeline. Per-region stage means discard the entire 246×246 connectivity structure, and they can only reveal effects large enough to survive averaging over 14–26 subjects per stage. The result does not show that the biomarkers carry no stage information; it shows that group means of single regions do not expose it.
4. The multivariate architecture is therefore motivated and not assumed: the transformer sees each subject individually and all 246 regions jointly, with connectivity intact.

No heatmap in this report is evidence of progression, and the pipeline has not been tested on any classification task.

Figure 6.2 places the DC_z and FC_Strength heatmaps side by side, which turns the redundancy of Section 5.6 into a visual proof.

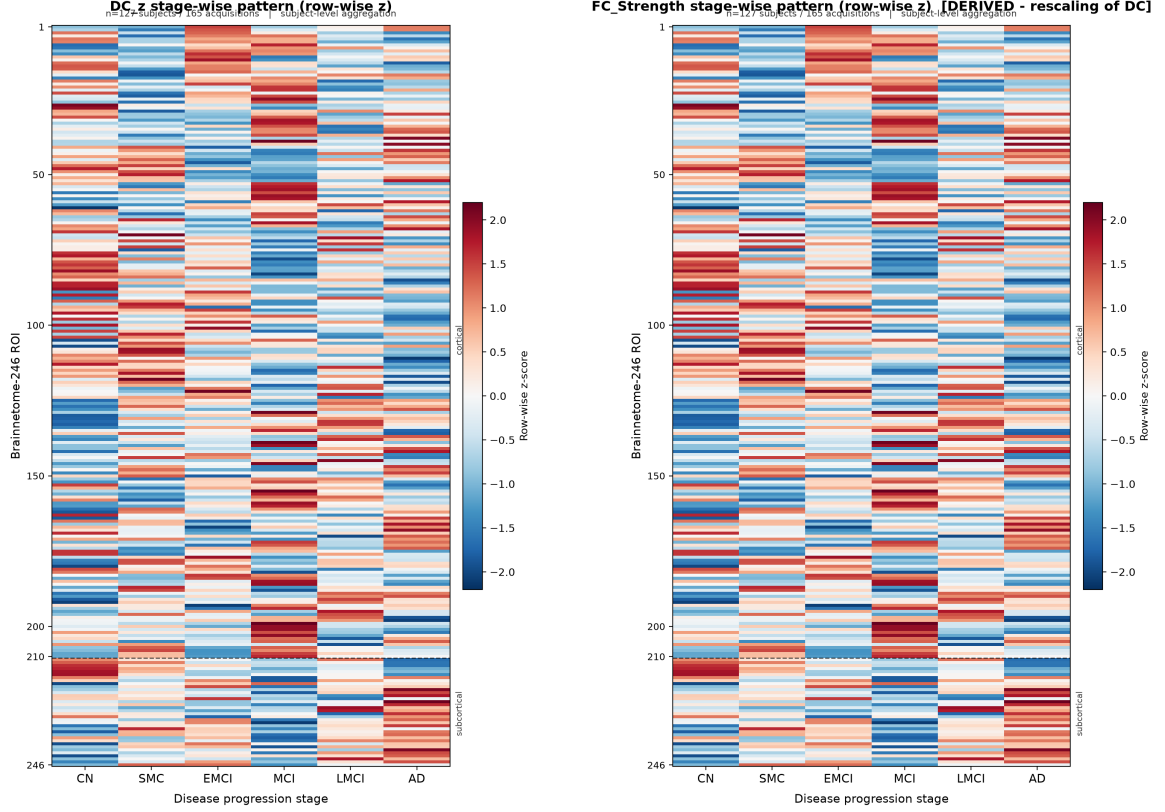


Figure 6.2: DC_z (left) and $FC_Strength$ (right), rendered independently from separate stored files. The heatmap panels are identical, because $FC_Strength = DC/245$ exactly (maximum difference 6.939×10^{-18} across all 165 acquisitions, Pearson $r = 1.000000000000$); the plotted row-wise z values differ by at most 6.4×10^{-15} . The right-hand file is wider only because its title carries a “derived” tag. $FC_Strength$ therefore carries no information independent of DC .

6.5 Work Not Yet Done

To avoid any ambiguity about the scope of this report, the following are *not* done.

- **Comorbidity data are not cleaned.** Five of the six cohort table archives are present; the LMCI tables are missing entirely, leaving 20 subjects without CDR, MMSE, GDS, medical-history or APOE4 data. Of the 21 subjects lacking APOE4, 20 are the whole LMCI group, so the missingness is group-structured, and an availability indicator would act as a proxy label for LMCI. It must not be used as a feature.
- **Hypertension and diabetes are not directly available.** The ADNI medical-history tables supply only the broad categories MH4CARD (cardiovascular) and MH9ENDO (endocrine); they are used under those names.
- **Modules 4–6 are not started:** the multi-attention transformer, meta-

learning, comorbidity fusion, the Bayesian ordinal classifier and the dashboard. No accuracy, AUC or other performance figure exists, and none is implied anywhere in this report.

- **Cross-validation is not built.** Subject-level folds and train-fold-only cross-subject scaling are still to be implemented.

6.6 Discussion

Comparison with conventional practice. The pipeline departs from a textbook rs-fMRI workflow at two points, in both cases because the data compels it. A conventional pipeline co-registers BOLD to a subject T1w image and normalises through it, and derives tissue regressors from that subject’s own segmentation. Neither is possible here, so EPI-to-template registration is performed directly and tissue regressors are taken from population priors. This is a genuine limitation rather than a simplification: direct EPI registration is less precise because EPI has lower anatomical contrast, and its quality cannot be benchmarked against a T1w-mediated alternative using this dataset. Equally, a conventional pipeline applies slice-timing correction; the exhaustive audit described in Chapter 5 establishes this is not available, and the honest response was to omit the step and document it rather than assume an acquisition order.

On scale and feature redundancy. Two findings will shape the modelling stage more than any preprocessing choice. First, raw ALFF varies by more than two orders of magnitude across acquisitions, because it is expressed in units proportional to raw BOLD intensity, whereas ReHo — being bounded — varies only narrowly. Supplying both unnormalised to a model would allow scanner and protocol scaling to dominate any biological signal; the within-subject normalisation of Module 5 addresses this. Second, one of the four nominal biomarkers was found to be an exact rescaling of another, so the independent regional feature count is three, not four. Both findings were identified by inspecting the data rather than inherited from the design, which illustrates the value of auditing features before modelling.

Limitations. Beyond the absence of T1w and slice-timing data, three limitations bear on the conclusions that can eventually be drawn. The effective sample is 127 subjects across six ordered classes with a smallest class of 14, which bounds model capacity and makes a single held-out test split uninformative; repeated subject-grouped cross-validation is required. Eight acquisitions are permanently excluded on account of the 6.02 s TR subgroup unless the filter specification is revisited for them. Finally, the comorbidity data are incomplete and the gap is group-structured; see Section 6.5, which also records what has not yet been done.

Next steps. Immediate work is to complete the comorbidity table, construct the subject-level cohort index with fixed stratified grouped cross-validation folds so that cross-subject scaling can be fitted on training data only, and then implement the multi-attention transformer and meta-learning blocks. The ordinal scale used by the classifier is the one fixed at Review I, namely $CN \rightarrow SMC \rightarrow EMCI \rightarrow MCI \rightarrow LMCI \rightarrow AD$, which orders the three mild-cognitive-impairment cohorts as early, intermediate and late. This ordering determines the loss structure of the ordinal model and is therefore held fixed throughout.

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